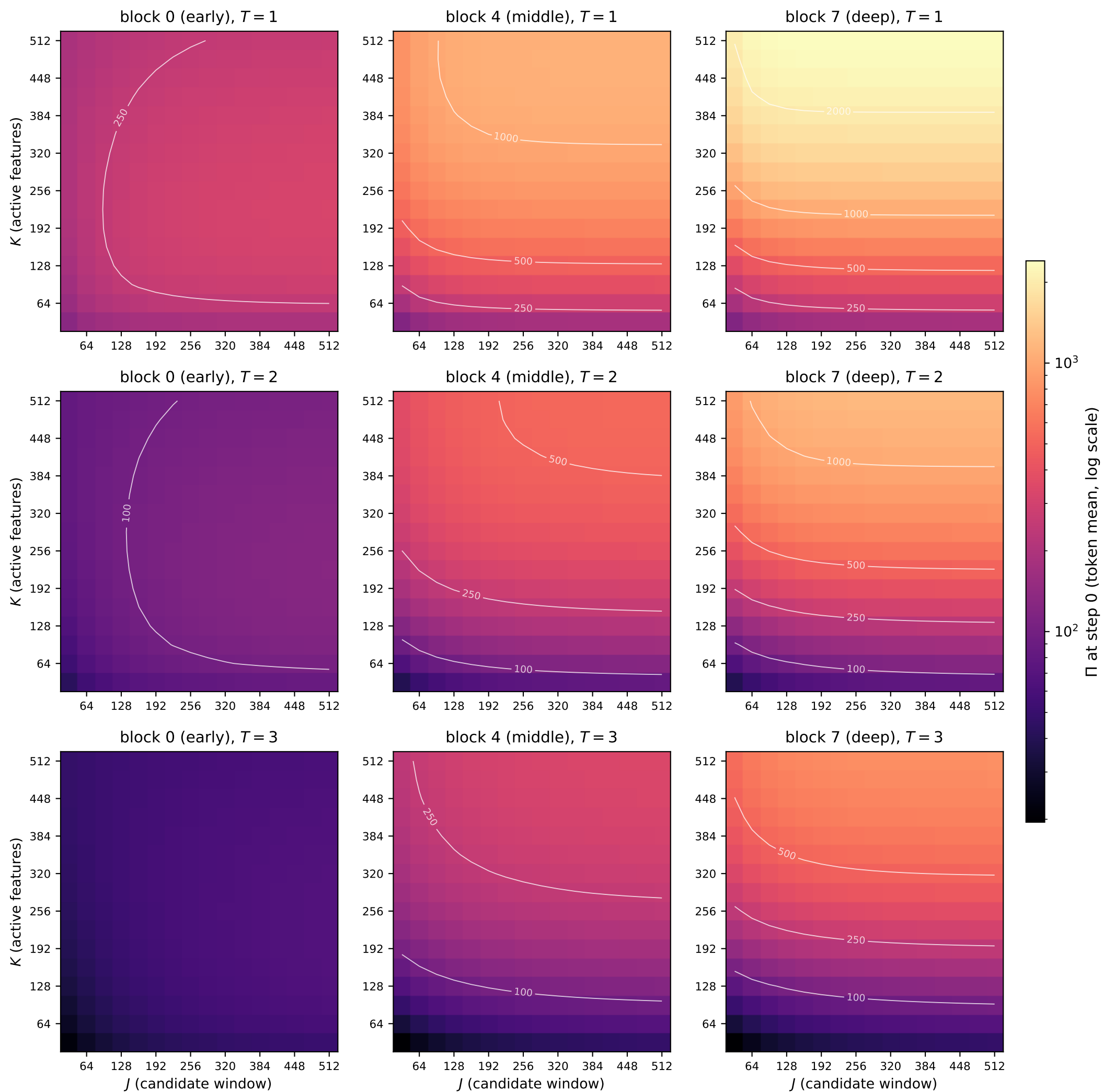
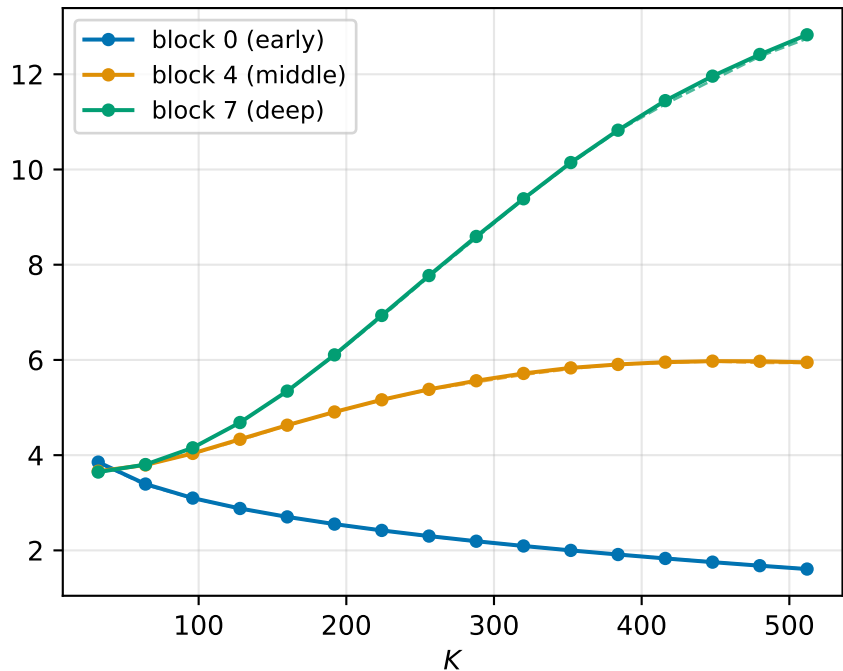


Surrogate gain $\Pi = \|L_\kappa D_z\|_F^2$ at initialization, rblapsum through_rank_kappa ($b_0 = 0$)

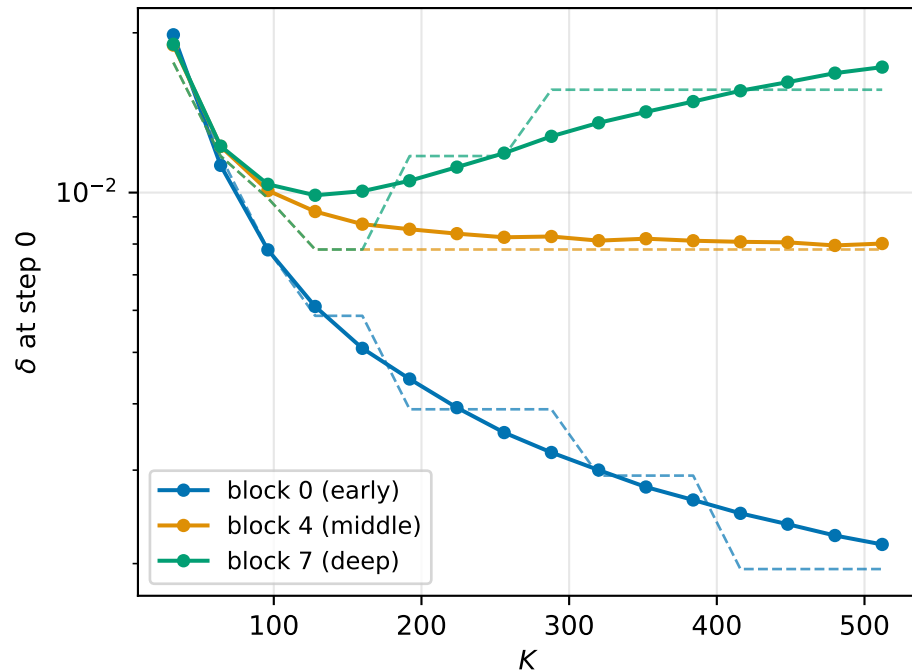


Boundary geometry at initialization (b and δ depend on K only, not on J or T)

boundary $b = s_{(K+1)}$ (solid mean, dashed median)



rank spacing $\delta = (s_{(K-4)} - s_{(K+4)})/8$



proxy Π_{approx}

